

FOREST FIRE DETECTION USING ARTIFICIAL INTELLIGENCE SOLUTIONS

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Abstract

Forest fires is a terrible occurrence which often causes the government billions of funds to repair the massive damage, it is a big threat to humans, wildlife, and the economic survival of every nation. Often developing countries don't survive the economical destruction caused by the forest fire. It is one of the most severe and dangerous forms of unanticipated disasters in the world. There have been other methodologies which were proposed and implemented in the detection of forest fire, some of which couldn't give the most accurate and early detection. In this study a support vector machine-based system for forest fire detection was proposed. The forest fire prediction system is based on the meteorological features of the input data corresponding to the critical weather features that influences forest fire occurrence i.e., temperature, relative humidity, windspeed and rain. Five machine learning models were trained on Algerian forest fire dataset from the uci repository, data pre-processing was done and further exploratory data analysis such as data cleaning, visualization were carried out on the dataset, although it was an imbalanced dataset, hence balanced using synthetic minority oversampling technique (SMOTE) where the minority category (not-fire) was oversampled to the proportion of the majority category(fire), statistical analysis was carried out to visualize the relationship between the independent variables and the output class using pairplots, histogram, correlation heatmap, box and whisker plot, and scatter plot. Feature selection was carried out to determine the most important features of the independent variables to the forest fire detection approach using SelectKBestMethod on the meteorological features of weather, the result was visualized using a bar plot that shows that Relative Humidity followed by Temperature have the highest effect on forest fire occurrence. Machine learning implementation was computed by training the models; logistic regression (89% test accuracy and 15ms wall time), decision tree classifier (96% test accuracy and 5ms wall time), random forest classifier (93% test accuracy and 350ms wall time), gradient Boosting classifier (93% test accuracy and 87ms wall time) and support vector machine (93% test accuracy and 2ms wall time). The algorithms were further evaluated and compared using RepeatedStratifiedKFold and cross validation score where $n_splits = 10$ and $n_repeats=3$. Logistic regression with mean accuracy of 83.5% and standard deviation of 0.084, decision tree classifier with mean accuracy of 89.3% and standard deviation of 0.062, random forest classifier with mean accuracy of 92.2% and standard deviation of 0.062, gradient boosting classifier with mean accuracy of 90.8% and standard deviation of 0.074, and support vector machine with mean accuracy of 91.8% and standard deviation of 0.074. The results were compared and visualized in a box and whisker plot which shows that random forest classifier and support vector machine are having almost equal performance on the average, this was subjected to statistical hypothesis test using P-value and t-statistic. Therefore, support vector machine

with overall accuracy of 91.8%, execution time of 2ms was deployed as a web-based application using Flask (a web framework) and Heroku cloud computing infrastructure which host the predictive model on an Heroku dynos container and performs real time forest fire prediction based on the four-weather input features; temperature, relative humidity, windspeed and rainfall. This is followed by continuous logging, production checks and performance evaluation of the model to ensure correct predictions. The results was compared to that of an imbalanced dataset and other existing work in forest fire detection machine learning domain.

Introduction

Forest is an integral part of the earth's ecology, which gives a lengthy list of support to both humans and wildlife, its destruction by fire continues to have a direct impact on the ecosystem and, the loss of life and property affects the economic stability of the world. (Borja et al., 2019). Forest fires are unplanned, uncontrollable disaster which cause the unrepairable destruction of natural vegetation (Borja et al., 2016). The destruction of the forest leads to serious environmental degradation such as global warming (Pan et al., 2013).

Aa a result of global warming on 19th July 2022, there was a forest fire incident in the London Wennington area following a temperature surge above 40 degrees Celsius causing the wide spread of fire from the grassland into residential homes destroying both properties and human lives. Late detection of forest fires causes major economic damage and loss of humans and wide lives (Borja et al, 2018), hence the need to develop an artificial intelligence and data science solutions as a machine learning model for early detection of forest fires.

The null hypothesis (H_0), states that there is no effect or no relationship between independent variables and the target class of the dataset. The alternative hypothesis (H_a), states that there is an effect or relationship between independent variables and the target class of the dataset. The aim and objectives of this study can be defined as the development of a support vector machine-based model with the goal being the model deployment as a part of a smart sensor node architecture system that gives accurate forest fire predictions at the early stage to prevent loss to property, wildlife, and humans in automated and intelligent way without human intervention.

2.0 Background

Forest fire smoke detection approach using MVMNet by (Hu et al., 2022), they collected images in different burning conditions and various climate conditions across forest areas, MVMnet outperforms all other methods.

Forest Fires Detection Using Machine Learning Techniques by (Elshewey and Elsonbaty, 2020). It is a regression task; linear, ridge, and lasso regression models were used in the detection of fire-prone areas. Two trainings approach where in one, the 13 features were used for training, while 70% were used in the second. Linear regression outperformed other models.

Forest fire detection using machine learning by (Pragati, Shambhuwani, and Umbrajkar, 2020). They trained a decision tree and support vector algorithms, the decision tree algorithm outperformed the support vector with the test accuracy of 99%, hence the decision tree algorithm was recommended for forest fire detection and reduction of false alarm and human constant supervision.

A review on forest fire detection techniques by (Alkhatib, 2014), the author reviewed a list of techniques used for forest fire detection; controlled burning, fire weather forecasts, watch towers, optical smoke detection, lightning detectors, and infrared.

Detection of Forest Fire Areas using Machine Learning by (Spoorthy and Kumar, 2022). They evaluated different models for quick forest fire detection; support vector machine, regression, decision tree, and K-Nearest Neighbour, the regression outperformed other models.

Predicting Forest Fire Using Remote Sensing Data and Machine Learning by (Yang et al., 2021). They evaluated and recommended a Neural Network that uses remote sensing data to predict forest fire.

A Data Mining Approach to Predict Forest Fires using Meteorological Data by (Cortez and Morais, 2007). In this paper, they evaluated various models, however, the support vector machine trained on 4 selected features; temperature, windspeed, relative humidity, and wind speed outperformed other models.

Artificial intelligence for forest fire prediction by (Sakr et al., 2010). They trained support vector algorithm on data from Lebanon using weather features for prediction, support vector algorithm outperformed other models.

Predicting Forest Fire in Algeria Using Data Mining Techniques: Case Study of the Decision Tree Algorithm (Abid and Izeboudjen, 2019). In this paper, they performed classification task using weka tool on meteorological data from two areas in Algeria, decision tree was recommended for integration into software architecture for forest fire detection.

Forest fire detection using machine learning by (Rajan and Paul, 2022). They trained and evaluated the performance of a convolutionary neural network (CNN) with 93% accuracy for forest fire detection.

Predicting Burned Areas of Forest Fires: An Artificial Intelligence Approach by (Castelli, Vanneschi, and Popovič, 2015). They proposed a geometric semantic genetic programming approach using 517 forest fire events between 2000 and 2003 as database, which outperformed when compared to standard genetic programming.

Predicting forest fires with different data mining techniques by (De, Labdhi, and Garg, 2020). In this paper They trained algorithms on forest fire dataset from uci repository; support vector machine, logistic regression, and random forest in a regression task for the prediction of burnt areas of forest fire.

Predicting Forest Fire Hazards Using Data Mining Techniques: Decision Tree and Neural Networks by (Hamadeh, Karouni, and Daya, 2014), In this paper They explored two data

mining approach; decision tree and back propagation Neural networks (Feed forward function fitting neural network and cascade feed forward neural network), the decision tree was trained on four weather attributes while the later was trained on two attributes, the back propagation forward neural network achieved the best performance, i.e., network with two inputs.

Real-time forest fire detection framework based on artificial Intelligence using color probability model and motion feature analysis by (Wahyono et al., 2022). In this paper They proposed and evaluated a new framework for forest fire detection using color-motion-shape features of the forest fire and machine learning algorithms for predictive modelling. The new framework achieved fastest and highest accuracy.

A review of early forest fire detection systems using optical remote sensing by (Barmpoutis et al., 2020), In this paper They reviewed the optical remote sensing technologies and the functionality of the smoke, flame algorithms used in forest fire detection, the systems are as follows; terrestrial, airborne, and spaceborne- systems.

3.0 METHODOLOGY

The data science and machine learning approach used in this project was explained in this section.

3.1 DATASET DESCRIPTION

Figure 1 FWI SYSTEM

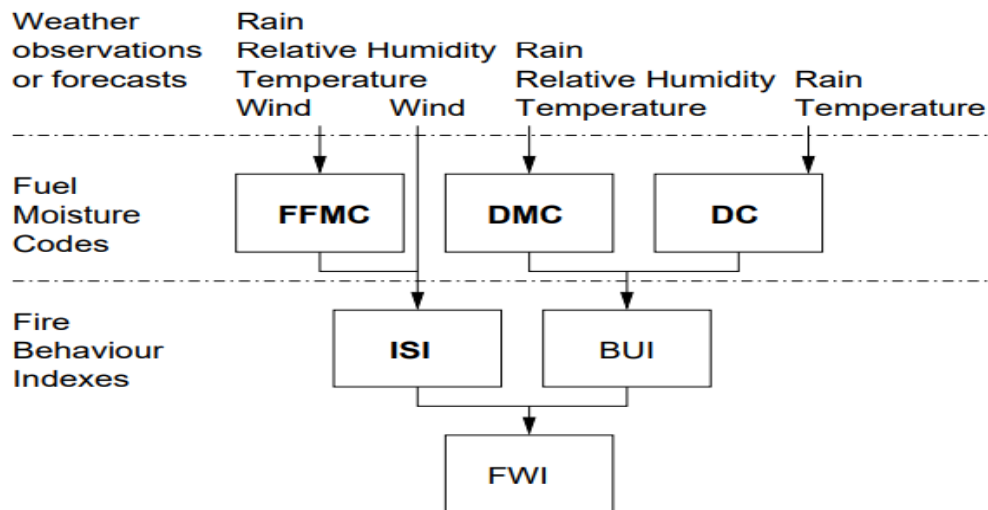


Table 1. Dataset Description

Dataset Characteristics	Multivariate	Number of Instances	244
Attribute Characteristics	Real	Number of Attributes	12
Associated Tasks	Classification & Regression	Missing Values	N/A

Date: Day, Month, and Year

Weather Conditions: Temperature, Relative Humidity (RH), Wind speed (Ws), and Rain.

FWI Components: Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC), Drought Code (DC), Initial Spread Index (ISI), Build-up Index (BUI), and Fire Weather Index (FWI).

Output Classes: Fire and not-fire.

3.2 EXPLORATORY DATA ANALYSIS

This sub-section provides the data analysis method used in preparing the dataset for machine learning classification, regression task, and further deployment of model into cloud server.

3.2.1 LOADING DATASET IN PANDAS DATAFRAME

It was observed that the dataset was not displayed properly in pandas, so the first row 'Bejaia Region Dataset' was skipped to get a better view of the dataframe using 'skiprows=1'. The data types were in object, having 246 entries and 14 columns.

3.2.2 MISSING VALUE ANALYSIS AND DATA CLEANING

The missing value count was computed, visualized using the missing value heatmap and the histogram plot which shows that there are missing values in all the columns except 'Day' and the output classes has the maximum counts. The dataset description shows that the dataset consists of 244 datapoints but Forest_Fire.shape shows 246 datapoints, hence there are 2 extra datapoint which needs dropping.

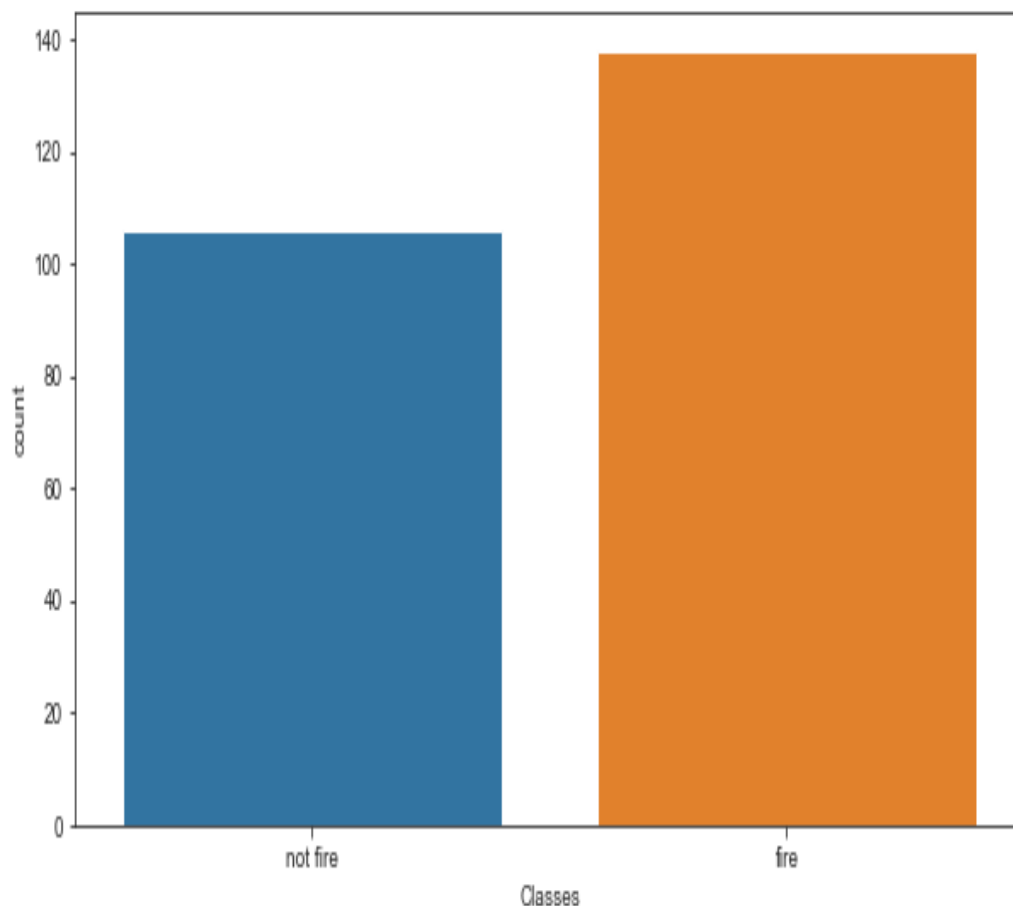
Estimating the data-points of the target variable for each class shows that some of the categories appears to be same, also the category named 'Classes' appears to be an irrelevant label, the spaces were replaced with '_' (hyphenated) symbol and all '_' (hyphenated) symbol were replaced with '' (empty string). The row of irrelevant label was dropped as it counts as one only, the empty spaces were removed from the features using strip ().

Also, entry error on the 'Day' column on the 122nd row starting with 'Sidi-Bel Abbes Region Dataset', the entry error on Drought code column on the 167th row having 14.6 and 9 below it, and the fire weather Index column on the 167th row having 'fire' category. The datapoints from the above observations was manually pre-processed. Further cleaning was computed by dropping 122nd row and Sidi-Bel Abbes Region dataset entry error. The 123rd row with repeated column headings was dropped.

3.3 EVALUATING IF DATASET IS BALANCED OR NOT

The percentage of fire is 57%, and the percentage of non-fire is 43%. The percentage of fire was subtracted from the percentage of not-fire in the target class of the dataset, and threshold set at 5%, if and else statement was utilized i.e., if the difference is over 5%, the dataset is imbalanced else balanced, but difference in the percentage of fire and non-fire is 14% which can be translated as an imbalanced dataset i.e., two categories of the target class having different distribution i.e., fire (1) as the majority category while not-fire (0) is the minority category.

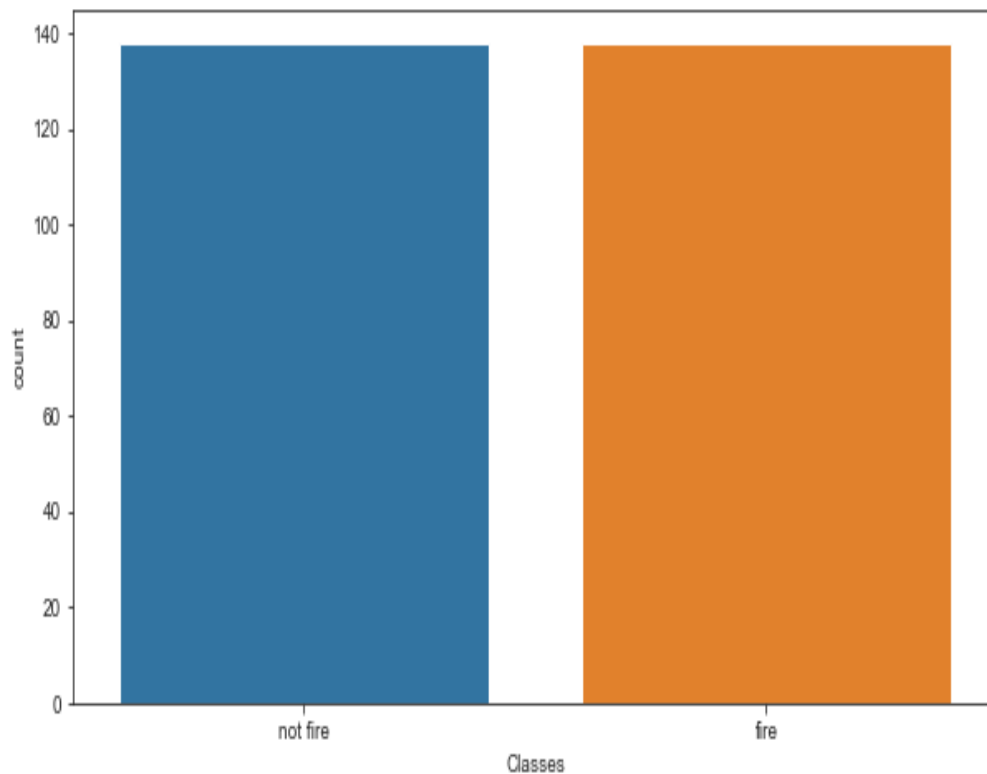
Figure 2 Imbalanced dataset



3.4 TARGET CLASS EXPLORATION

The minority category of the output class i.e., not-fire with less distribution was oversampled by adding synthetic distribution to add up the difference with the majority category i.e., fire (1) using synthetic minority oversampling technique (SMOT). The count plot below shows fire and not-fire having equal distribution.

Figure 3. Countplot of balanced dataset



As of result of oversampling, the new dataframe with 276 entries and 14 columns was produced, furthermore the output class was encoded, where 0 as "not-fire" and 1 as "fire". The categorical variables were converted to numerical variables i.e., object type to integer and float type.

3.5 BRIEF OVERVIEW OF ALGORITHMS AND THE MATHEMATICAL EQUATIONS

- Logistic Regression Algorithm are supervised machine learning algorithm for binary classification of dependent variables.

Equation 1 Logistic function

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Equation 2 Best line of fit in linear regression

$$y = \beta_0 + \beta_1 x$$

Equation 3 Odds of probabilities

$$P = \beta_0 + \beta_1 x$$

$$\frac{P}{1-P} = \beta_0 + \beta_1 x$$

Equation 4 Log of odds

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x$$

Equation 5 Probabilities

$$p = \frac{e^{(\beta_0 + \beta_1 x)}}{1 + e^{(\beta_0 + \beta_1 x)}}$$

$$1 = \frac{e^{(\beta_0 + \beta_1 x)}}{p} - e^{(\beta_0 + \beta_1 x)}$$

$$p[1 + e^{(\beta_0 + \beta_1 x)}] = e^{(\beta_0 + \beta_1 x)}$$

$$p = \frac{e^{(\beta_0 + \beta_1 x)}}{1 + e^{(\beta_0 + \beta_1 x)}}$$

- Decision Tree Classifier are supervised machine learning algorithm used for regression and classification task.

Equation 6

$$Entropy = - \sum_{i=1}^n p_i * \log(p_i)$$

Equation 7

$$Gini\ index = 1 - \sum_{i=1}^n p_i^2$$

- Random Forest Classifier are supervised machine learning model used for regression and classification task, achieves satisfactory performance with less parameter tuning.

Equation 8 Gini index

$$\begin{aligned} Gini\ Index &= 1 - \sum_{i=1}^n (P_i)^2 \\ &= 1 - [(P_+)^2 + (P_-)^2] \end{aligned}$$

Equation 9 Entropy

$$E(S) = -p_{(+)} \log p_{(+)} - p_{(-)} \log p_{(-)}$$

Equation 10

$$f_i = \frac{\sum_{j: \text{node } j \text{ splits on feature } i} n_j}{\sum_{k \in \text{all nodes}} n_k}$$

- Gradient boosting is a form of the ensemble approach that involves many less strong decision trees and added together for better performance.

Equation 11 Initialize model with a constant value

$$F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^n L(y_i, \gamma)$$

Equation 12 Compute residue

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{m-1}(x)} \quad \text{for } i = 1, \dots, n$$

Equation 13 Compute γ_m

$$\gamma_{jm} = \underset{\gamma}{\operatorname{argmin}} \sum_{x_i \in R_{jm}} L(y_i, F_{m-1}(x_i) + \gamma) \text{ for } j = 1, \dots, J_m$$

Equation 14 Update the model

$$F_m(x) = F_{m-1}(x) + v \sum_{j=1}^{J_m} \gamma_{jm} 1(x \in R_{jm})$$

- Support Vector Machine are supervised machine learning model which takes an array of data and sort it into two categories in classification task.

Equation 15 Hinge loss function

$$c(x, y, f(x)) = \begin{cases} 0, & \text{if } y * f(x) \geq 1 \\ 1 - y * f(x), & \text{else} \end{cases}$$

Equation 16 loss function for support vector machine

$$\min_w \lambda \|w\|^2 + \sum_{i=1}^n (1 - y_i \langle x_i, w \rangle)_+$$

Equation 17 Gradients

$$\begin{aligned} \frac{\delta}{\delta w_k} \lambda \|w\|^2 &= 2\lambda w_k \\ \frac{\delta}{\delta w_k} (1 - y_i \langle x_i, w \rangle)_+ &= \begin{cases} 0, & \text{if } y_i \langle x_i, w \rangle \geq 1 \\ -y_i x_{ik}, & \text{else} \end{cases} \end{aligned}$$

Equation 18 Gradient update (no misclassification)

$$w = w - \alpha \cdot (2\lambda w)$$

Equation 19 Gradients Update (misclassification)

$$w = w + \alpha \cdot (y_i \cdot x_i - 2\lambda w)$$

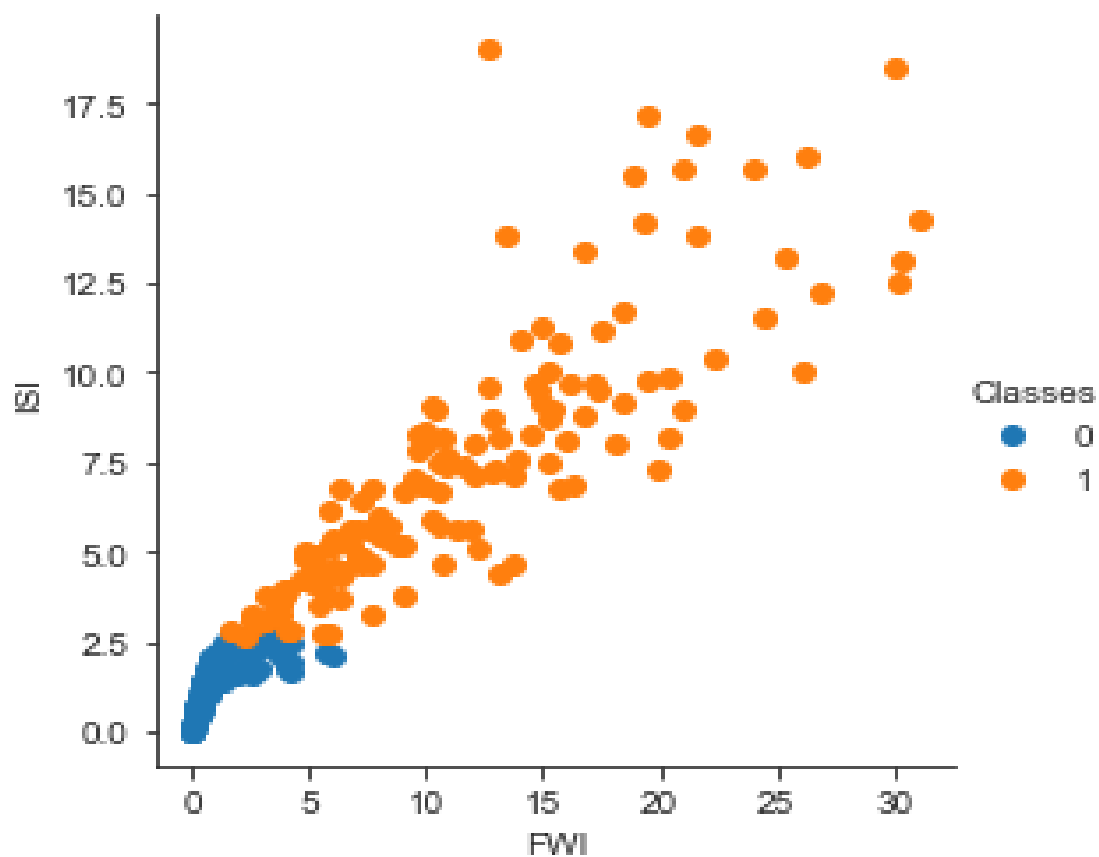
4.0 RESULTS

This section covers the quantitative and statistical evaluation of independent variables and the output class, model training, performance testing and evaluation of the classification algorithms and the deployment of support vector machine into development and production environment (Heroku cloud computing server).

4.1 QUANTITATIVE AND STATISTICAL EVALUATION OF VARIABLES

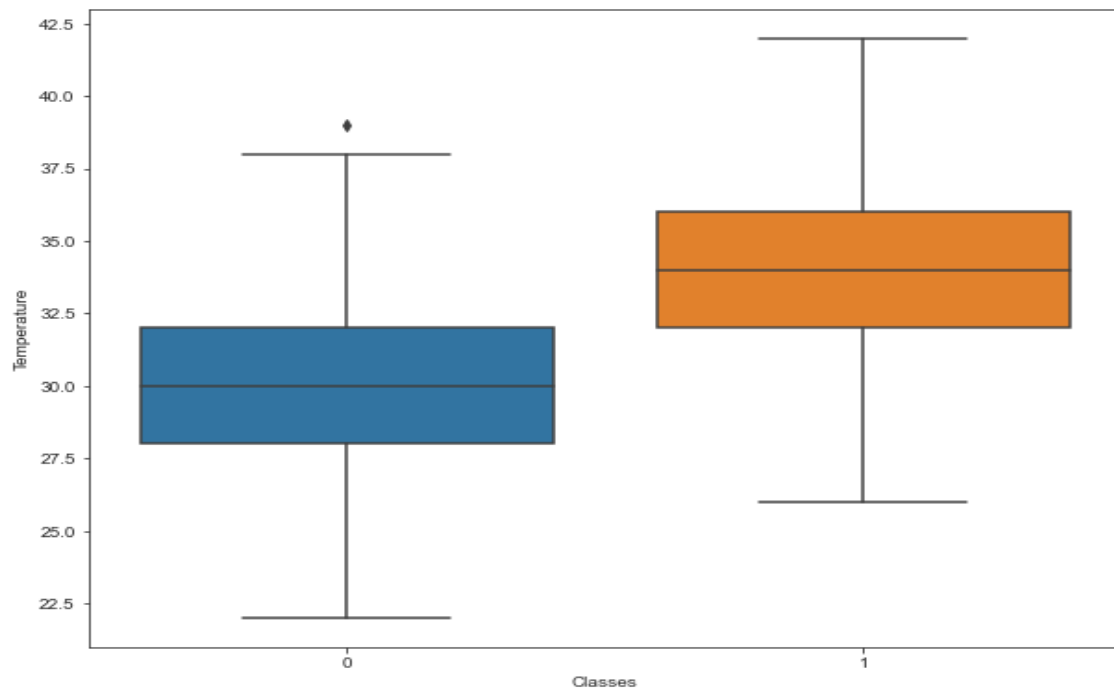
The scatter plots show the relationships between variables, i.e., output class, Fire Weather index and Initial Spread Index.

Figure 4 Scatterplot of the Output class, FWI, ISI



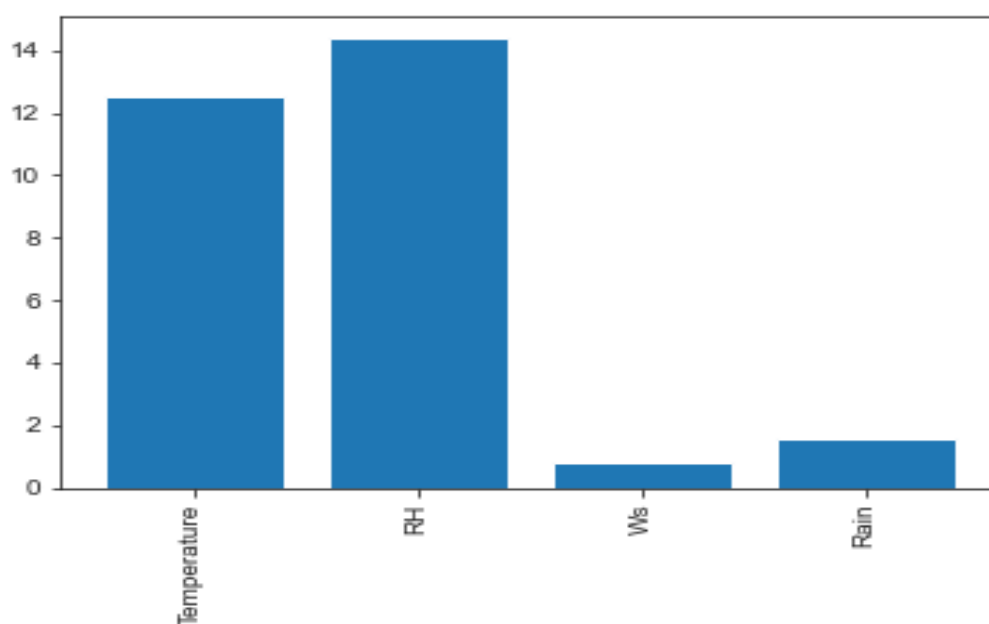
The boxplot shows the distribution/skewness of predictive classes fire (1) and not-fire (0) in the Temperature Feature.

Figure 5. The boxplot of Temperature and Classes



Feature selection using selectkbest method, the bar plots shows that relative humidity and temperature strongly influences forest fire.

Figure 6 bar plot of selected predictors/features



4.2 MODEL TRAINING

The dataset was transformed using standard scaler, a feature scaling approach to ensure all features have same magnitude and dimension. The training dataset was fitted to the machine learning models to learn and make predictions.

4.3 COMPARING CLASSIFIER ALGORITHMS USING REPEATEDSTRATIFIEDKFOLD AND CROSS_VAL_SCORE.

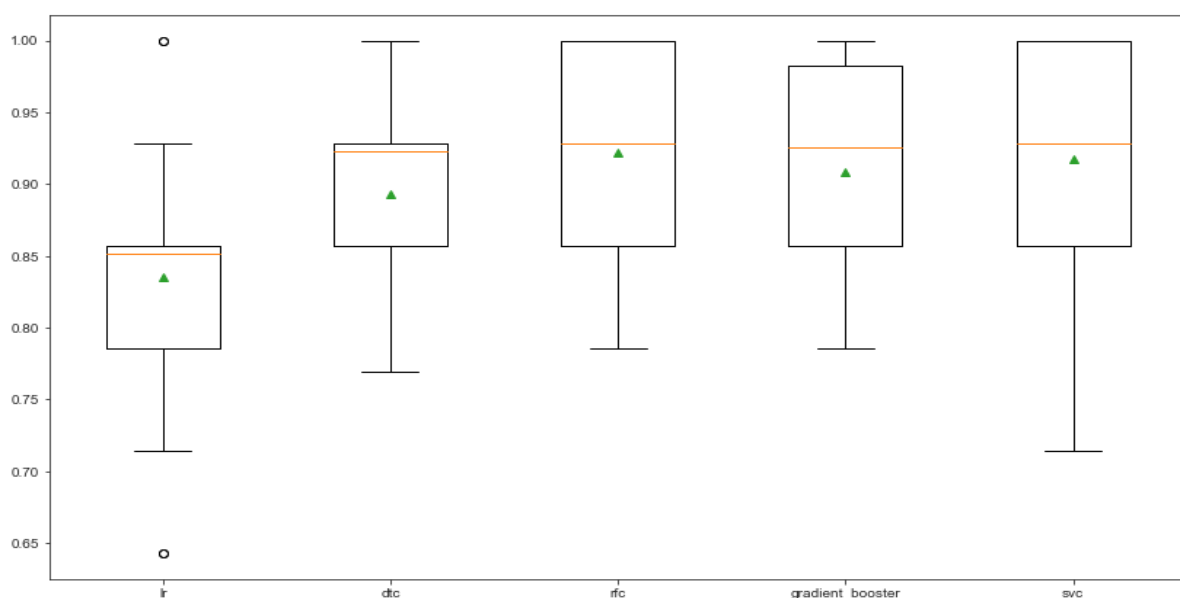
Table 2 Comparing Classifier Algorithms using RepeatedStratifiedKFold and cross_val_score

Given that n_splits=10, n_repeats=3

Classifier Algorithms	Mean Accuracy	Standard deviation
Logistic Regression	0.835	0.084
Decision Tree	0.893	0.062
Random Forest	0.922	0.062
Gradient Boosting	0.908	0.074
Support Vector Machine	0.918	0.074

Box and whisker plot shows performance comparison of the algorithms

Figure 7 box and whisker plot of the Algorithms



4.3 STATISTICAL HYPOTHESIS TEST.

To check if the result is statistically significant, the difference between random forest classifier and support vector machine mean accuracy was evaluated, where p-value = 0.717, and t-Statistic = 0.384, hence if p-value ≤ 0.05 , the difference between the algorithms is probably real else algorithms probably have the same performance. Whereas 0.717 is > 0.05 , hence algorithms have same performance, therefore support vector machine and random forest classifier will have the same performance on the average. Null hypothesis accepted as there is no significant difference between the two algorithms.

4.4 PERFORMANCE COMPARISON

Table 3 Performance comparison for the 'fire' class

Algorithms	Wall Time (ms)	Precision	Recall	f1-score	Support	Testing Accuracy	Mean Accuracy	Standard Deviations
Logistic Regression	15	0.67	1.00	0.80	6	0.89	0.835	0.084
Decision Tree	5	1.00	0.83	0.91	6	0.96	0.893	0.062
Random Forest	350	1.00	0.67	0.80	6	0.93	0.922	0.062
Gradient Boosting	87	1.00	0.67	0.80	6	0.93	0.908	0.074
Support Vector Machine	2	1.00	0.67	0.80	6	0.93	0.918	0.074

Table 4 Performance comparison for the 'not-fire' class

Algorithms	Wall Time (ms)	Precision	Recall	f1-score	Support	Testing Accuracy	Mean Accuracy	Standard Deviations
Logistic Regression	15	1.00	0.86	0.93	22	0.89	0.835	0.084
Decision Tree	5	0.96	1.00	0.98	22	0.96	0.893	0.062
Random Forest	350	0.92	1.00	0.96	22	0.93	0.922	0.062
Gradient Boosting	87	0.92	1.00	0.96	22	0.93	0.908	0.074
Support Vector Machine	2	0.92	1.00	0.96	22	0.93	0.918	0.074

4.5 COMPARISON WITH THE EXISTING WORK

Table 5 Comparison with the Existing work

References	Model	Accuracy %	Recall	F-measure
This study	Support Vector machine	91.8	0.67 & 1.00	0.80 & 0.96
This study	Decision tree	89.3	0.83 & 1.00	0.91 & 0.98
Stojanova et al., 2006	Decision tree	81.2	0.81	0.81
Sakr et al., 2010	Support Vector machine	87.72	Not given	Not given
Abid et al., 2020	Decision Tree	82.89	0.92	0.85

4.6 RESULTS FROM MODELLING ON IMBALANCED DATASET

This section provides the result from modelling and evaluation done on an imbalanced dataset.

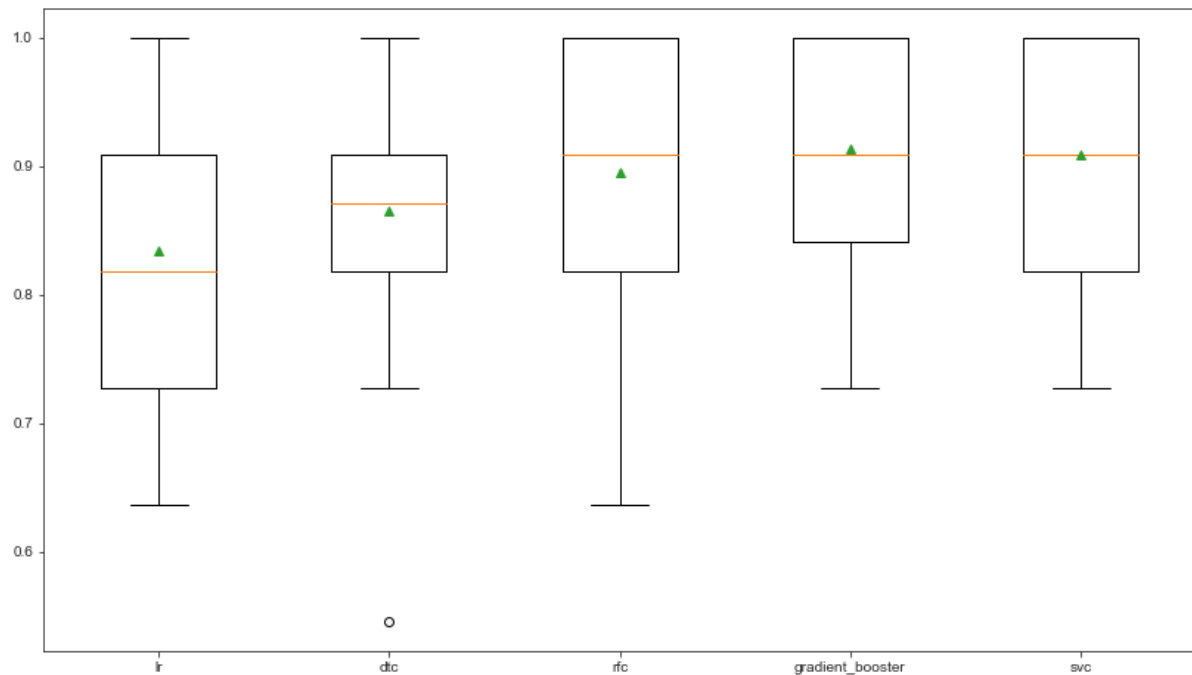
4.7 COMPARING CLASSIFIER ALGORITHMS USING REPEATEDSTRATIFIEDKFOLD AND CROSS_VAL_SCORE.

Table 6 Comparing Classifier Algorithms using RepeatedStratifiedKFold and cross_val_score

Classifier Algorithms	Mean Accuracy	Standard deviation
Logistic Regression	0.834	0.110
Decision Tree	0.865	0.096
Random Forest	0.894	0.105
Gradient Boosting	0.913	0.083
Support Vector Machine	0.909	0.094

Box and whisker plot shows performance comparison of the algorithms.

Figure 8 box and whisker plot of the Algorithms



4.7.1 STATISTICAL HYPOTHESIS TEST.

As P-value 0.114 is > 0.05 , hence algorithms have same performance, therefore support vector machine and gradient boosting Classifier will have the same performance on the average.

4.8 PERFORMANCE COMPARISON OF IMBALANCED DATASET

Table 7 Performance comparison for the 'fire' class

Algorithms	Wall Time (ms)	Precision	Recall	f1-score	Support	Testing Accuracy	Mean Accuracy	Standard Deviations
Logistic Regression	14	0.57	0.80	0.67	5	0.83	0.834	0.110
Decision Tree	4	0.75	0.60	0.67	5	0.87	0.865	0.096
Random Forest	393	0.80	0.80	0.80	5	0.91	0.894	0.105
Gradient Boosting	59	0.67	0.40	0.50	5	0.83	0.913	0.083
Support Vector Machine	3	0.80	0.80	0.80	5	0.91	0.909	0.094

Table 8 Performance comparison for the 'not-fire' class

Algorithms	Wall Time (ms)	Precision	Recall	f1-score	Support	Testing Accuracy	Mean Accuracy	Standard Deviations
Logistic Regression	14	0.94	0.83	0.88	18	0.83	0.834	0.110
Decision Tree	4	0.89	0.94	0.92	18	0.87	0.865	0.096
Random Forest	393	0.94	0.94	0.94	18	0.91	0.894	0.105
Gradient Boosting	59	0.85	0.94	0.89	18	0.83	0.913	0.083
Support Vector Machine	3	0.94	0.94	0.94	18	0.91	0.909	0.094

4.6 SUPPORT VECTOR MACHINE DEPLOYMENT

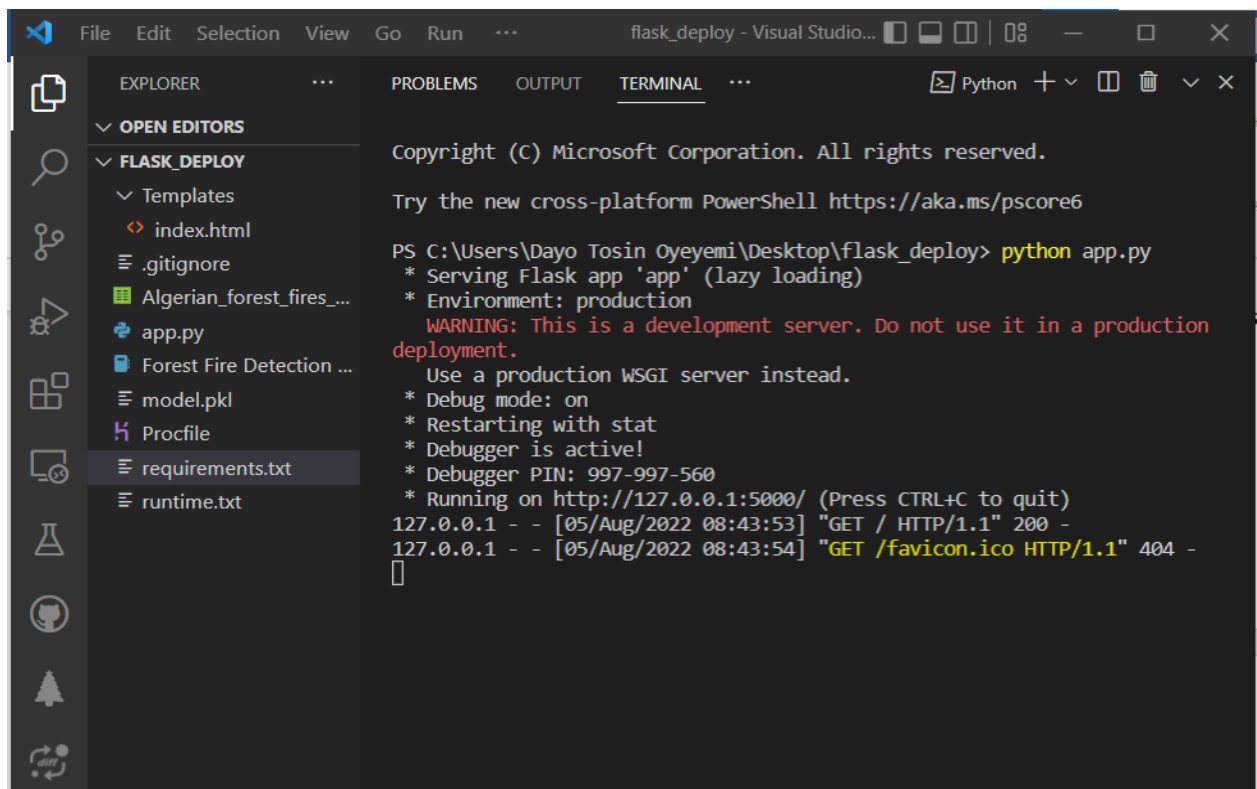
The machine learning model deployment life cycle was explained in this section. The model was saved by dumping support vector machine in the python pickle library, support vector machine was deployed into production(Heroku cloud computing infrastructure server) using Flask micro-web framework by creating a code pipeline which connects GitHub repository holding the development files with the Heroku production infrastructure where the application is hosted remotely on the free dynos which serves as a container for the web-based application and performs real time prediction for up to 1000hours monthly, when the dynos does not receive a web request in a 30-minute period, it sleeps to save up hosting hours. Application slug size of 142.1MiB of 500MiB and Python/Heroku as the buildpacks

The application main directory consists of the following files.

- i. Templates folder which holds file named index.html which serves as frontend of the application made up of bootstrap, css and html codes. Basically, it serves as the homepage of the application.
- ii. gitignore file holds an extensive list of untracked files which are to be ignored during the production process by Git
- iii. The csv file holds the Algerian forest fire dataset.
- iv. app.py file holds the flask application, loaded model from the pickle library and backend codes, it is the application root.
- v. Jupiter notebook file holds the Exploratory Data analysis, Cleaned and balanced dataset, visualization, trained models, and the saved model.
- vi. model.pkl file holds the dumped model, which enables objects to be serialized to files on disk and deserialized back into the program at runtime.
- vii. Procfile which provides the web process type during the production, it specifies the commands that are executed by the application on the startup.
- viii. requirements.txt file listing all the dependencies for the web-based application development.
- ix. Runtime.txt contains the python version for the application development.

All the files listed were pushed to GitHub repository from which a code pipeline connecting the development files to the Heroku cloud server was created.

Figure 9 Development and testing environment



- **Links to application**

The links provide access to the web-based application for Real time forest fire prediction

<https://flask--prediction.herokuapp.com/>

This link is a customized domain name which provides access to the web-based application for real time forest fire Prediction although not secured (without SSL certificate) because of the free Heroku Dynos.

<http://forestfirepredictor.com/>

4.6.1 List of Technology used

- Jupiter notebook
- Pandas
- Matplotlib
- Seaborn
- SMOTE
- Scikit-learn
- Python
- Git
- Github

- Visual Studio Code
- Heroku
- Flask
- Pickle
- CSS
- HTML
- Bootstrap
- Gunicorn
- Click
- Numpy
- Sklearn
- Werkzeug
- Jinja2
- Colorama

5.0 DISCUSSION

5.1 PERFORMANCE EVALUATION METRICS

Wall Time (ms): This is the model training time i.e., execution time of the classifier algorithm, it takes support vector machine 2ms and decision tree algorithms 5ms to train/learn as seen on table 3, which are the fastest of the other classifier algorithms.

Precision: A machine learning model without false positives has a precision of 1.0, from table 3, for the fire class (1), logistic Regression produces false positives hence precision of 0.67 while decision tree classifier, random Forest classifier, gradient boosting classifier, and support vector machine have no false positives hence precision equals to 1.0. Hence when support vector machine predicts fire (1) it is correct 100% of the time.

However, table 4, for the not- fire class (0), logistic regression has no false positives hence precision equals to 1.0, however the other classifiers have false positives hence precision of 0.96, and 0.92, respectively. Hence when support vector machine predicts not-fire (0) it is correct 92% of the time.

Equation 20

$$\text{Precision} = \frac{TP}{TP + FP}$$

TP= True positive, FP= False positive.

Recall: A machine learning model without false negatives has a recall of 1.0, from table 3 of the fire class (1) Logistic regression has no false negatives hence recall equals to 1.0, however decision tree classifier, random forest classifier, gradient boosting, and support vector machine have false negative hence recall equals to 0.83, and 0.67 respectively. Hence when support vector machine identifies fire (1) it is 67% correct of all the time.

From table 4 of the not-fire class (0), decision tree classifier, random forest classifier, gradient boosting classifier and support vector machine have no false negative hence recall equals 1.0 but logistic regression has false negative, hence recall equals 0.86. Hence when support vector machine identifies not-fire (0) it is 100% correct of all the time.

Equation 21

$$\text{Recall} = \frac{TP}{TP + FN}$$

TP= True positive, FN= False negative.

F1-score: The F-measure, often known as the F-score, is a measurement of a test's accuracy used in statistical analyses of binary categorization. It is derived from the test's recall and precision. An F-score can have a maximum value of 1.0, which denotes perfect precision and recall, and a minimum value of 0, which occurs when either precision or recall are zero.

Equation 22 f1 score

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{tp}{tp + \frac{1}{2}(fp + fn)}$$

The harmonic mean of recall and precision is known as the classic F-measure or balanced F-score (F1 score). Hence from table 3 and 4 the F-score for support vector machine is 0.80 & 0.96 for fire (1) and not-fire (0) class respectively meaning the precision and recall for fire (1) is 0.80 while the precision and recall for not-fire (0) is 0.96

Support

The number of samples of the genuine answer that fall into each class of target values can be used to figure out support. The sum of 28 samples falls into the testing set, hence for support vector machine and other classifier algorithms in table 3 and 4, for the fire class (1) there were 6 samples of true response while 22 samples of true response for the not-fire (0) class.

Testing Accuracy: Testing Accuracy is the percentage of correctly predicted outcomes for the test data

Equation 23 Testing Accuracy

$$\text{accuracy} = \frac{\text{correct predictions}}{\text{all predictions}}$$

Decision tree classifier has the highest test accuracy of 96%, logistic regression with the least test accuracy of 89% while random forest classifier, gradient boosting classifier, and support vector machine have equal test accuracy of 93%.

Mean Accuracy: This is the overall accuracy computed using RepeatedStratifiedKFold and cross_val_score where number of split equals to 10 and repeated 3 times for each classifier algorithm, as a result of this 30 different accuracies were derived for each algorithm, the mean accuracies and standard deviations were calculated hence support vector machine and random forest have nearly equal overall accuracies of 0.918 and 0.922 while logistic regression, decision tree classifier, and gradient boosting classifier have accuracies of 0.835, 0.893, and 0.908 respectively.

Standard Deviations: values were computed following the RepeatedStratifiedKFold and cross_val_score for the classifiers i.e., logistic regression, decision tree, random forest, gradient boosting, and support vector machine; 0.084, 0.062, 0.062, 0.074, 0.074. The standard deviation can gauge how confident one is in the statistical findings of a model. Using the standard deviation, one can easily spot outliers. A low standard deviation shows that the data are tightly grouped around the mean (more dependable), while a high standard deviation shows that the data are widely dispersed (less dependable).

5.2 PERFORMANCE COMPARISON OF BALANCED AND IMBALANCED DATASET USING SUPPORT VECTOR MACHINE AS A CASE STUDY.

- Testing accuracy declined from 93% to 91%
- Wall time Increased from 2ms to 3ms
- Overall accuracy declined from 91.8% to 90.9%
- Standard deviation Increased from 0.074 to 0.094

For fire classification

- Precision declined from 1.0 to 0.80
- Recall increased from 0.67 to 0.80
- F1-score remained constant

For not-fire classification

- Precision increased from 0.92 to 0.94
- Recall decreased from 1.00 to 0.94
- F1-score declined from 0.96 to 0.94

CONCLUSION

This project presents the data analysis, data science, machine learning and software engineering approach used in the data collection, exploration, modelling, statistical evaluation, development and deployment of a predictive support vector machine-based model as a web-based application, using Flask web framework into the Heroku cloud computing infrastructure server for real time forest fire prediction based on the four meteorology features of weather; temperature, relative humidity, windspeed and rain. Forest fire is a disastrous occurrence to the ecosystem, early detection of it will preserve, safeguard lives, and protect properties against fire destruction. This system is a simple artificial Intelligence-based solution that will not only save both lives and properties but also alert firefighters about forest fire behaviour with meteorological weather features.

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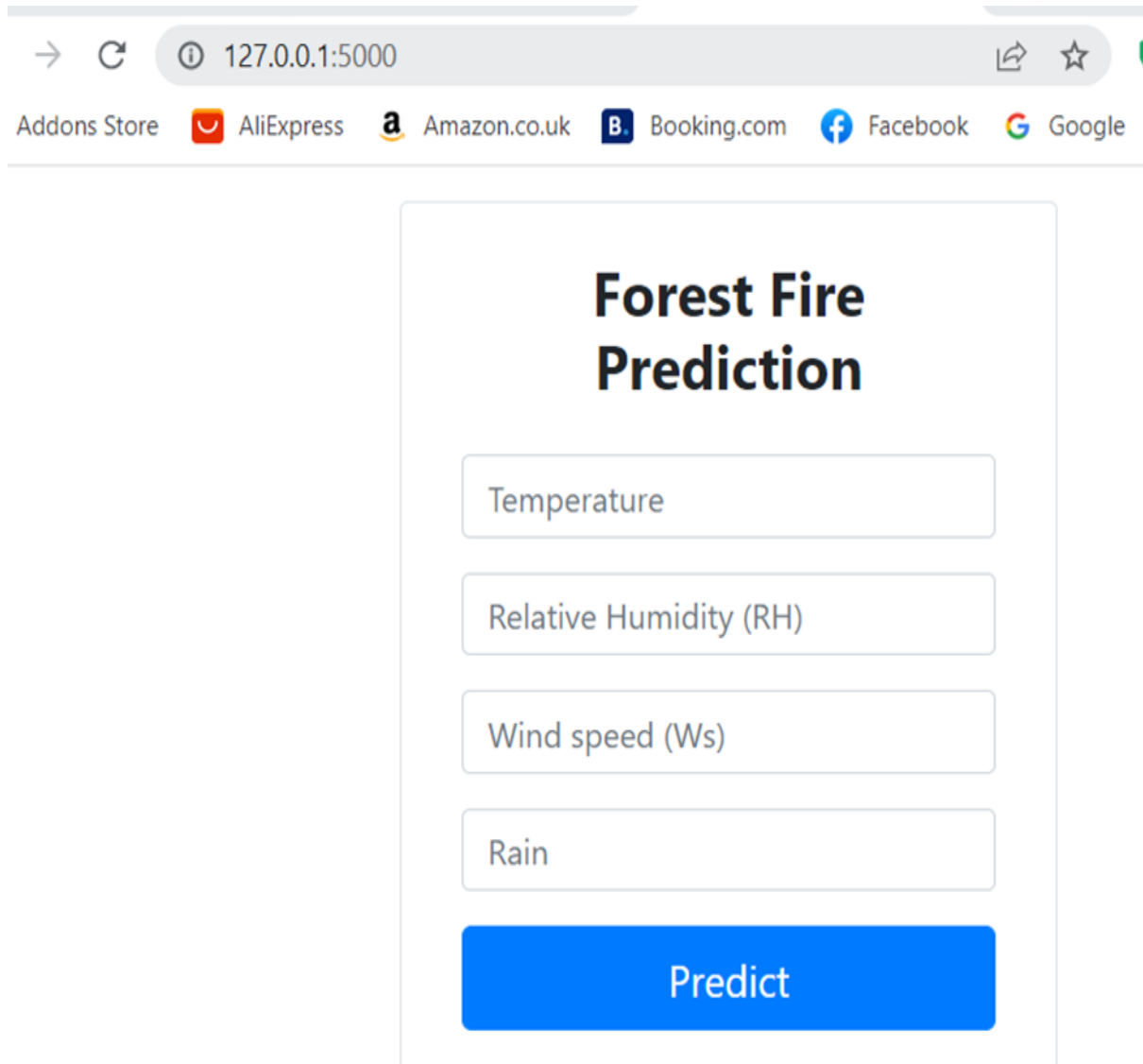
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APPENDIX

Figure 10 Testing Development in web browser



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000". Below the address bar is a row of bookmarks including "Addons Store", "AliExpress", "Amazon.co.uk", "Booking.com", "Facebook", and "Google". The main content area features a "Forest Fire Prediction" form. The form has a title "Forest Fire Prediction" in bold black text. Below the title are four input fields: "Temperature", "Relative Humidity (RH)", "Wind speed (Ws)", and "Rain". At the bottom of the form is a blue button labeled "Predict".

→ ↻ ⓘ 127.0.0.1:5000

Addons Store AliExpress Amazon.co.uk Booking.com Facebook Google

Forest Fire Prediction

Temperature

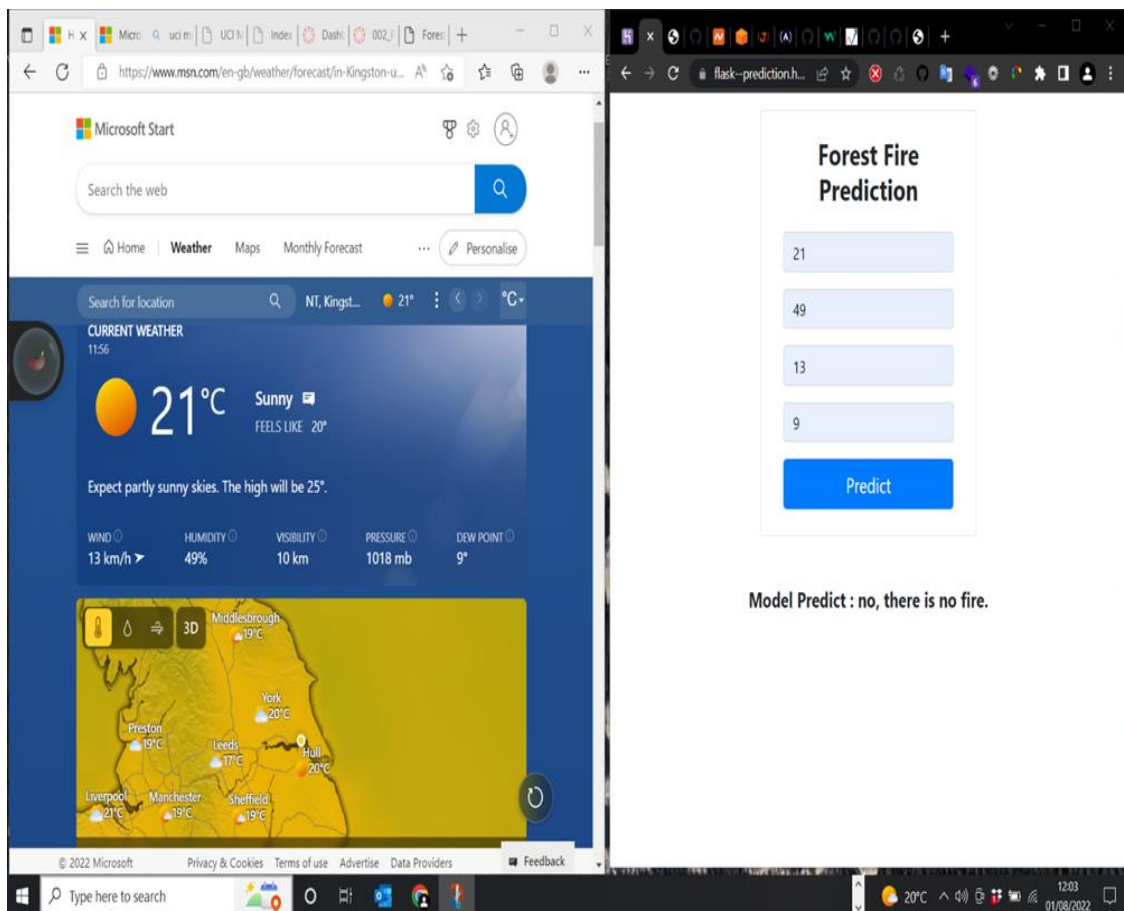
Relative Humidity (RH)

Wind speed (Ws)

Rain

Predict

Figure 11 Forest Fire Real time prediction



Algorithms classification Report

Logistic Regression

	precision	recall	f1-score	support
0	1.00	0.86	0.93	22
1	0.67	1.00	0.80	6
accuracy			0.89	28
macro avg	0.83	0.93	0.86	28
weighted avg	0.93	0.89	0.90	28

Decision Tree Classifier

	precision	recall	f1-score	support
0	0.96	1.00	0.98	22
1	1.00	0.83	0.91	6
accuracy			0.96	28
macro avg	0.98	0.92	0.94	28
weighted avg	0.97	0.96	0.96	28

Random Forest Classifier

	precision	recall	f1-score	support
0	0.92	1.00	0.96	22
1	1.00	0.67	0.80	6
accuracy			0.93	28
macro avg	0.96	0.83	0.88	28
weighted avg	0.93	0.93	0.92	28

Gradient Boosting Classifier

	precision	recall	f1-score	support
0	0.92	1.00	0.96	22
1	1.00	0.67	0.80	6
accuracy			0.93	28
macro avg	0.96	0.83	0.88	28
weighted avg	0.93	0.93	0.92	28

Support Vector Machine

	precision	recall	f1-score	support
0	0.92	1.00	0.96	22
1	1.00	0.67	0.80	6
accuracy			0.93	28
macro avg	0.96	0.83	0.88	28
weighted avg	0.93	0.93	0.92	28